MILITARY SPECIFICATION

GENERATOR, HYDROGEN (HYDROGEN GENERATOR ML–303( )/TM)

1. SCOPE

1.1 This specification covers one type of portable, manually operated generator for generating hydrogen gas by the reaction of water on calcium hydride, designated as Hydrogen Generator ML–303( )/TM. (See 6.2.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ–S–571	—Solder, Lead Alloy, Tin Lead Alloy, and Tin Alloy, Flux Cored Ribbon and Wire, and Solid Form
QQ–S–781	—Steel Strapping, Flat
UU–T–111	—Tape, Paper, Gummed (Sealing and Securing)
PPP–B–566	—Boxes, Folding, Paperboard
PPP–B–621	—Boxes, Wood, Nailed and Lock-Corner
PPP–B–636	—Box, Fiberboard
PPP–P–291	—Paperboard, Wrapping, Cushioning
PPP–T–76	—Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing)

MILITARY

MIL–P–116	—Preservation, Methods of
MIL–B–117	—Bags, Interior Packaging
MIL–E–1409	—Envelope, Pay Record
MIL–M–13231	—Marking of Electronic Items
MIL–F–14072	—Finishes for Ground Signal Equipment

STANDARDS

MILITARY

MIL–STD–105	—Sampling Procedures and Tables for Inspection by Attributes
MIL–STD–129	—Marking for Shipment and Storage

DRAWINGS

ELECTRONICS COMMAND

SC–D–7431	—Hydrogen Generator MIL–303/TM Assembly and Details
SC–A–7433	—Punch, Assembly and Details
SC–DL–87908	—Hydrogen Generator ML–303( )/TM
SC–GL–51535	—Gages for Hydrogen Generator ML–303/TM

(Copies of specifications, standard, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained

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from the procuring activity or as directed by the contracting officer. Both the title and number of symbol should be stipulated when requesting copies.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 **Description.** Hydrogen Generator MIL–303( )/TM is a portable, manually operated generator for generating hydrogen gas in the field. The hydrogen is generated by the reaction of water on calcium hydride in charge containers attached to the generator by interrupted threads. A punch is included with the hydrogen generator for punching out the knockouts in the top of the charge container.

3.2 **Preproduction samples.** The contractor shall furnish five preproduction samples for approval, if required by the invitation for bids and contract. (See 4.3.)

3.3 **Construction.** The equipment shall be constructed in accordance with Drawing and Data List SC–DL–87908.

3.4 **Cleaning.**

3.4.1 *Parts.* After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled into the equipment.

3.4.2 *Units.* After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5 **Finish.** The equipment shall be finished in accordance with Specification MIL–F–14072 and the equipment drawings. (See 4.4.)

3.6 **Marking.**

3.6.1 *General.* Marking shall conform to Specification MIL–M–13231 and the applicable equipment drawings. (See 4.4.)

3.6.2 *Serial numbers.* Serial numbers under Specification MIL–M–13231 are not required.

3.7 **Soldering.**

3.7.1 *Solder.* Solder shall conform to Specification QQ–S–571 and the equipment drawing.

3.7.2 *Acid or acid salts.* Acid or acid salts may be used in preparation for, and during soldering, including preliminary tinning and soldering of mechanical joints. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.7.3 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, or under heated. The minimum necessary amount of flux and solder shall be used. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment.

3.8 **Seams.** All external seams shall be gastight and shall be capable of meeting the leakage test of 4.9.

3.9 **Punch.** A punch in accordance with Drawing SC–A–7433 shall be provided with each Hydrogen Generator ML–303( )/TM.

3.10 **Weight.** The weight of the complete Hydrogen Generator ML–303( )/TM (including punch) shall not exceed 2 pounds.

3.11 **Interchangeability.** Like units and replaceable parts shall be physically and

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functionally interchangeable, without modification of such items or of the equipment. (See 4.8). Individual items shall not be hand-picked for fit or performance.

3.12 Technical literature, tools, and running spare parts. Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order.

3.13 Preconditioning. The equipment shall be capable of meeting the requirements herein without subsequent processing, after subjection to the bounce preconditioning of 4.6. (Also see 4.5.)

3.14 Workmanship. The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein.

3.4 Cleaning
3.5 Finish
3.6 Marking
3.7.2 Flux and cleaning agents for soldering
3.7.3 Process for soldering
3.8 Seams

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3.)
(b) Inspection covered by subsidiary documents. (See 4.4.)
(c) Quality conformance inspection.
  (1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5.)
  (2) Quality conformance inspection of preparation for delivery. (See 4.10.)

4.3 Preproduction inspection. This inspection will be performed by the Government *unless otherwise specified in the contract.* It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A and group B.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	*Where required*
Finish	3.5
Marking	*3.6*

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4, 4.5.1, and 4.5.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government—at its discretion—may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.3.) Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to group A or group B inspection, except preproduction samples, shall be preconditioned after final assembly. (See 3.13.)

4.5.1 *Group A inspection.* This inspection, including sampling, shall conform to table I

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and the ordinary inspection procedures of Standard MIL–STD–105. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE I. *Group A Inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL	
			Major	Minor
Visual and mechanical.	.....	4.9	1%	4%
Leakage.....	3.8	4.7	1%	

4.5.2 *Group B inspection.* This inspection, including sampling, shall conform to paragraph 4.8 and to the special procedures for small-sample inspection of Standard MIL–STD–105. The AQL shall be 1 percent defective and the inspection level shall be L–7 for normal and tightened inspection and L–5 for reduced inspection. The reduced inspection procedure shall be R–1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.6 Bounce preconditioning. The unit shall be placed in its normal operation position on the table of the Package Tester, Type 1000–SC, as made by the L.A.B. Corporation, Skaneateles, N.Y., or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.7 Leakage test. Hydrogen Generator ML–303( )/TM shall be completely submerged in water for 15 seconds, with the outlet tube capped, so as to trap air in the body of the generator. There shall be no evidence of leakage of air from the generator assembly.

4.8 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) The bottom of the generator body shall be tested to determine that the interrupted threads are compatible with the gages listed on Gage List SC–GL–51535.

(b) Outlet tubes and screwneck threads shall be measured for conformance to Drawing SC–D–7431.

4.9 Visual and mechanical inspection. The equipment shall be examined for defects listed in table II.

TABLE II. *Classification of Visual and Mechanical Defects*

Classification	Defects
Major	Insufficient solder. Solder does not completely fill or cover seams where intended. Cold solder joint. Chalky appearance, lacks metallic luster, presents rough "pile up" appearance; movement of solder upon pick application. Seams not properly filled with rubber compound where intended. Gasket missing. Rough or sharp edges on seams. Marking not in accordance with drawing, markings missing or obliterated.
Minor	Bare metal showing through finish. Solder not flowed to a feather edge.

4.10 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with Specification MIL–P–116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.

5.1.1 *Level A.* Hydrogen Generator ML–

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303( )/TM shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in Specification MIL–P–116, and as described in the following paragraphs.

5.1.1.1 *Technical manual.* Package each Technical Manual Method IC–6 as follows: Inclose each Technical Manual within a close-fitting, Type II, Class d bag conforming to Specification MIL–B–117. Heat seal to effect closure.

5.1.1.2 *Prepackaging.* Place and firmly secure the outlet tube gasket, required for initial operation, inside the outlet tube. Secure one each punch to the body cylinder by means of water-resistant, pressure-sensitive tape, conforming to Specification PPP–T–76.

5.1.1.3 *Outlet tube (spare).* Package each outlet tube Method III as follows: Cushion each tube by wrapping in Type I, paperboard, conforming to Specification PPP–P–291. Secure cushioning by means of Class 1, gummed paper tape conforming to Specification UU–T–111. Place one each cushioned tube within a close-fitting, Type II, Class A, folding carton conforming to Specification PPP–B–566. Seal closure in accordance with the requirements of the box specification.

5.1.1.4 *Body cylinder.* No packaging required. Secure the outlet tube, required for initial operation to the body cylinder.

5.1.1.5 *Gaskets, outlet tube (spares).* Package gaskets Method III as follows: Inclose spare gaskets within a close-fitting envelope conforming to Specification MIL–E–1409. Close envelope by stapling.

5.1.1.6 *Consolidated package.* Consolidate the package items specified in 5.1.1.1 through 5.1.1.5 together within a close-fitting Style RSC, Type I, class 2, W5c, fiberboard box conforming to PPP–B–636. Fill all voids with pads fabricated of cushioning material conforming to Class 2, of Specification PPP–P–291. Closure shall be in accordance with the appendix of the box specification.

5.1.2 *Level C.* Hydrogen Generator ML–303( )/TM shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing.

5.2.1 *Level A.* Place four (4) packaged Hydrogen Generator ML–303( )/TM within a close-fitting, Style 4, Class 2, nailed wood box conforming to Specification PPP–B–621. Closure shall be in accordance with the appendix of the box specification.

5.2.1.1 *Metal strapping.* Metal strapping, Types I, Class b, conforming to Specification QQ–S–781 shall be applied in accordance with the appendix of the box specification.

5.2.2 *Level B.* Hydrogen Generator ML–303( )/TM shall be packed for shipment as specified in paragraph 5.2.1 except the nailed wood box shall be Class I.

5.2.3 *Level C.* A quantity of Hydrogen Generators ML–303( )/TM shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Military Standard MIL–STD–129.

5.4 Package testing. Interior packages will be tested to assure compliance with the requirements specified herein.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

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(c) The specific paragraphs of Section 5 which are applicable to the particular procurement.
(d) Marking and shipping of samples.
(e) If rough handling test or cyclic exposure test and function test are required, so state.
(f) Place of final inspection.
(g) Technical literature required. (See 3.12.)
(h) Quantity of tools and running spare parts required. (See 3.12.)

6.2 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: ML–303W/TM. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.3 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in the specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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